

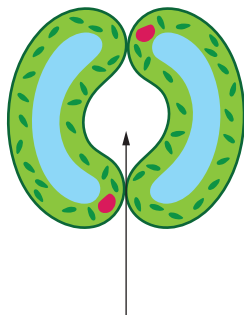
8. Photosynthesis is the process by which plants make food from water and carbon dioxide.

(a) Stomata are an important part of the leaf. In many plants, stomata remain open during the day and closed at night.

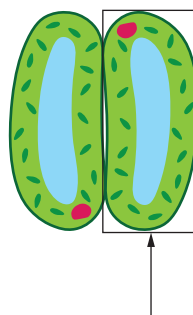
(i) Label the diagram below.

[2]

Open by day



Closed by night



(ii) State **two** functions of stomata.

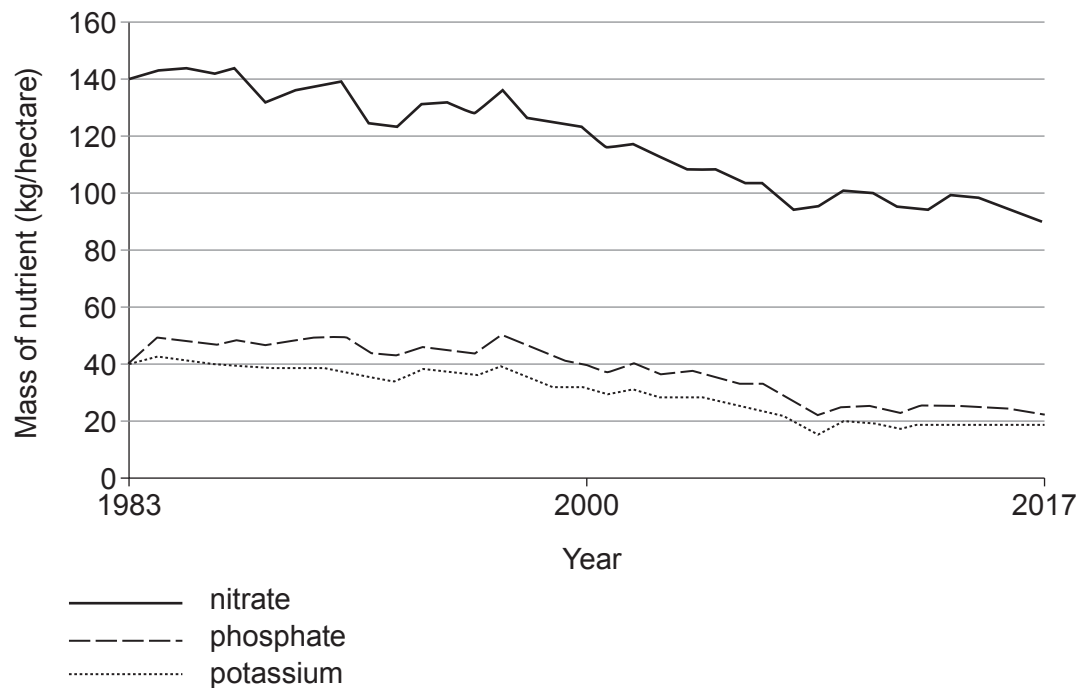
[2]

1.
2.



- (b) Healthy plants require a supply of nutrients. Many farmers use artificial fertilisers to supply these nutrients to the soil.

The graph shows the mass of different nutrients applied to agricultural land between 1983 and 2017.



Use the data from the graph to answer the following questions.

- (i) It is claimed that the use of nitrate fertilisers halved between 1983 and 2017. Use the graph to explain whether you agree.

[3]

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- (ii) The total land area of Wales is approximately 2 100 000 hectares.
60% of this land is used for farming.

I. Calculate the area of land used for farming in Wales.

[1]

area = hectares

II. Calculate the total mass of phosphates used in farming in Wales in 1983. [2]

mass = kg

END OF PAPER

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